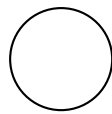


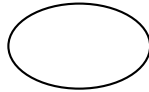
Identifying Conics and Reading Their Key Features

Naming a conic from its equation, and reading centres, radii, vertices, foci and asymptotes

- The four shapes below are the four conics. No values are shown: each problem supplies its own numbers.
- The sign test: with both squared terms on one side, let A and C be the coefficients of x^2 and y^2 . If $A \cdot C > 0$ the curve closes up (circle when $A = C$, otherwise an ellipse); if $A \cdot C < 0$ it opens in two branches (hyperbola); if one squared term is missing, it is a parabola.
- Keep exact values — radii and focal distances stay as whole numbers or surds, never decimals.



circle



ellipse



parabola



hyperbola

Circle: $(x - h)^2 + (y - k)^2 = r^2$

Parabola: $y = a(x - h)^2 + k$

Ellipse: $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$

Hyperbola: $\frac{(y - k)^2}{a^2} - \frac{(x - h)^2}{b^2} = 1$

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1. The equation $x^2 + y^2 = 49$ describes a circle centred at the origin. Find its radius.

Answer: _____

2. Consider $9x^2 + 4y^2 = 36$.

(a) Multiply the coefficient of x^2 by the coefficient of y^2 , and write the product.

Answer: _____

(b) Name the conic, and justify the name from what you found in part (a).

Answer: _____

3. A circle has a diameter whose endpoints are $(2, 1)$ and $(8, 9)$.

(a) Find the centre of the circle.

Answer: _____

(b) Find the radius of the circle.

Answer: _____

4. The parabola $y = x^2 - 6x + 5$ opens upward.

(a) Find the x -coordinate of its vertex.

Answer: _____

(b) Find the y -coordinate of its vertex.

Answer: _____

5. Consider $x^2 - 4y^2 = 16$.

(a) Multiply the coefficient of x^2 by the coefficient of y^2 , and write the product.

Answer: _____

(b) Name the conic, and justify the name from what you found in part (a).

Answer: _____

6. The ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ is centred at the origin.

(a) Find the length of the semi-major axis, that is, the distance from the centre to a vertex along the longer axis.

Answer: _____

(b) Find the distance from the centre to each focus.

Answer: _____

7. The parabola $y = x^2 + 8x + 11$ is to be written in vertex form.

(a) Find the x -coordinate of its vertex.

Answer: _____

(b) Find the y -coordinate of its vertex.

Answer: _____

8. The hyperbola $\frac{y^2}{36} - \frac{x^2}{64} = 1$ is centred at the origin and opens up and down.

(a) Find the distance from the centre to each vertex.

Answer: _____

(b) Find the slope of the asymptote that rises from left to right.

Answer: _____

9. The equation $x^2 + y^2 - 10x + 6y + 18 = 0$ hides a circle. Completing the square in x and again in y brings it into standard form.

(a) Find the centre of the circle.

Answer: _____

(b) Find the radius of the circle.

Answer: _____

10. The line $y = x + 1$ cuts across the circle $x^2 + y^2 = 25$.

(a) Find every point where the line meets the circle.

Answer: _____

(b) Explain how the number of real solutions tells you whether a line is a secant, a tangent, or misses the circle altogether.

Answer: _____